

Linear Algebra Review — Solution Manual

Duke StatSci Ph.D. Bootcamp · instructor copy

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Generated from *linear-algebra-bootcamp.qmd* — 20 problems. Regenerate with *python3 slides/make-solutions.py*.

Vector Spaces

1. Bases of a plane

slide 10 · Baseline · Companion review, Q15

Problem.

Find a basis for the plane $x - 2y + 3z = 0$ in $\mathbb{R}^3$. Then find a basis for the intersection of that plane with the xy -plane. Finally, find a basis for all vectors perpendicular to the plane.

Solution.

Write $x = 2y - 3z$, so a general point is $y(2, 1, 0)^\top + z(-3, 0, 1)^\top$: a basis is $\{(2, 1, 0)^\top, (-3, 0, 1)^\top\}$, and $\dim = 2$.

Intersect with $z = 0$: the single vector $(2, 1, 0)^\top$ spans a line.

The perpendicular direction is the coefficient vector itself, $(1, -2, 3)^\top$. Check: $\langle (1, -2, 3), (2, 1, 0) \rangle = 2 - 2 + 0 = 0$ and $\langle (1, -2, 3), (-3, 0, 1) \rangle = -3 + 0 + 3 = 0$. ✓

Moral: the coefficients of a linear equation *are* the normal vector. A hyperplane in $\mathbb{R}^p$ is the orthogonal complement of its coefficient vector.

2. Sums of subspaces

slide 11 · Proof stretch · Companion review, Q6

Problem.

Let S, T be subspaces of V and define $S + T := \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in S, \mathbf{t} \in T\}$.

- Prove $S + T$ is a subspace.
- If S and T are distinct lines through the origin in $\mathbb{R}^d$, what is the difference between $S \cup T$ and $S + T$? Prove $\text{span}(S \cup T) = S + T$.

Solution.

- $\mathbf{0} = \mathbf{0} + \mathbf{0} \in S + T$. For closure, $(\mathbf{s}_1 + \mathbf{t}_1) + a(\mathbf{s}_2 + \mathbf{t}_2) = (\mathbf{s}_1 + a\mathbf{s}_2) + (\mathbf{t}_1 + a\mathbf{t}_2)$, and each bracket stays in its own subspace.
- $S \cup T$ is two lines — **not** closed under addition, so not a subspace. $S + T$ is the plane they span. For the identity: $S \cup T \subseteq S + T$ and $S + T$ is a subspace, so $\text{span}(S \cup T) \subseteq S + T$; conversely any $\mathbf{s} + \mathbf{t}$ is a combination of an element of S and an element of T , hence lies in $\text{span}(S \cup T)$.

Linear Maps and Matrices

3. Building transformation matrices

slide 15 · Baseline · Companion review, Q1

Problem.

Find the 3×3 matrices acting on $\mathbf{x} \in \mathbb{R}^3$ that

- (a) project vectors onto the xy -plane;
- (b) reflect every vector about the xy -plane.

Solution.

Use the column rule: figure out where $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ go.

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Sanity checks worth internalizing now: $\mathbf{P}^2 = \mathbf{P}$ (projections are idempotent), $\mathbf{R}^2 = \mathbf{I}$ (reflections are involutions), and $\mathbf{R} = \mathbf{I} - 2(\mathbf{I} - \mathbf{P}) = 2\mathbf{P} - \mathbf{I}$.

4. Conformability

slides 16–17 · Baseline · Companion review, Q2 / Stage 1, Q2

Problem.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & -2 \end{bmatrix}$$
$$\mathbf{C} = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 1 & -1 \end{bmatrix} \quad \mathbf{d} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$$

- (a) Which of $\mathbf{AB}$, $\mathbf{BA}$, $\mathbf{ABC}$, $\mathbf{ABd}$, $\mathbf{BC}$, $\mathbf{BC}^\top$, $\mathbf{B}^\top \mathbf{C}$, $\mathbf{dC}$ are defined?
- (b) Compute $\mathbf{A}^\top(3\mathbf{B} + \mathbf{C})$.

Solution.

- (a) Defined: $\mathbf{AB}$ (2×3), $\mathbf{ABd}$ (2×1), $\mathbf{BC}^\top$ (2×2), $\mathbf{B}^\top \mathbf{C}$ (3×3). Not defined: $\mathbf{BA}$, $\mathbf{ABC}$, $\mathbf{BC}$, $\mathbf{dC}$.
- (b) Parentheses first: $3\mathbf{B}$ and $\mathbf{C}$ are both 2×3 , so the sum is defined. Then $\mathbf{A}^\top$ is 2×2 , and $(2 \times 2)(2 \times 3)$ gives a 2×3 .

$$3\mathbf{B} + \mathbf{C} = \begin{bmatrix} 3 & 0 & 6 \\ 9 & 3 & -6 \end{bmatrix} + \begin{bmatrix} 1 & -2 & 3 \\ -2 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 9 \\ 7 & 4 & -7 \end{bmatrix}$$
$$\mathbf{A}^\top(3\mathbf{B} + \mathbf{C}) = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & -2 & 9 \\ 7 & 4 & -7 \end{bmatrix} = \begin{bmatrix} 25 & 10 & -12 \\ 15 & 0 & 11 \end{bmatrix}$$

5. Dimension bookkeeping

slide 20 · Baseline · Companion review, Q8

Problem.

A 57×37 matrix $\mathbf{A}$ has rank 31. Give the dimensions of all four fundamental subspaces: $\text{Rank}(\mathbf{A})$, $\text{Rank}(\mathbf{A}^\top)$, $\text{Null}(\mathbf{A})$, $\text{Null}(\mathbf{A}^\top)$.

Solution.

$$\text{Rank}(\mathbf{A}) = \text{Rank}(\mathbf{A}^\top) = 31.$$

$$\text{Null}(\mathbf{A}) = 37 - 31 = 6 \text{ (apply rank-nullity to the 37 columns).}$$

$$\text{Null}(\mathbf{A}^\top) = 57 - 31 = 26 \text{ (apply it to } \mathbf{A}^\top, \text{ which has 57 columns).}$$

6. Null space of a stacked matrix

slide 21 · Bridge · Companion review, Q7

Problem.

Let $\mathbf{A} \in \mathbb{R}^{n \times d}$, $\mathbf{B} \in \mathbb{R}^{m \times d}$, and $\mathbf{C} = \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix} \in \mathbb{R}^{(m+n) \times d}$. State the relationship between $\ker(\mathbf{C})$, $\ker(\mathbf{A})$ and $\ker(\mathbf{B})$, and justify it.

Solution.

$$\ker(\mathbf{C}) = \ker(\mathbf{A}) \cap \ker(\mathbf{B}).$$

$$\mathbf{C}\mathbf{x} = \begin{bmatrix} \mathbf{A}\mathbf{x} \\ \mathbf{B}\mathbf{x} \end{bmatrix}, \text{ and a stacked vector is zero iff both blocks are zero.}$$

7. Elimination, pivots, back-substitution

slide 23 · Baseline · Companion review, Q4

Problem.

In the system below, what multiple of the first equation should be subtracted from the second? Convert to matrix form, run Gaussian elimination, identify the pivots, and back-substitute.

$$2x + 3y = 5, \quad 6x + 15y = 12$$

Solution.

Multiplier $\ell_{21} = 6/2 = 3$. Eliminating gives $6y = -3$, so $y = -\frac{1}{2}$, then $2x = 5 - 3(-\frac{1}{2}) = \frac{13}{2}$, so $x = \frac{13}{4}$.

Pivots: 2 and 6. Two pivots $\Rightarrow$ rank 2 $\Rightarrow$ unique solution.

8. Row reduction and null space

slides 24–25 · Bridge · Companion review, Q5

Problem.

$$\mathbf{A} = \begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$$

(a) Find the row-reduced form. (b) What is $\text{Rank}(\mathbf{A})$? (c) Find a basis for $\ker(\mathbf{A})$.

Solution.

$R_3 \leftarrow R_3 - 2R_1$ gives $(0, -12, -3, -21)$, which is $-3 \times (0, 4, 1, 7) = -3R_2$. So the third row is dependent:

$$\mathbf{A} \longrightarrow \begin{bmatrix} 1 & 0 & 23/4 & 1/4 \\ 0 & 1 & 1/4 & 7/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(b) Two pivots, so $\text{Rank}(\mathbf{A}) = 2$.

(c) Free variables x_3, x_4 ; nullity $= 4 - 2 = 2$. A basis:

$$(-23, -1, 4, 0)^\top, \quad (-1, -7, 0, 4)^\top.$$

(Clear the fractions by scaling — always allowed for a basis.)

Inner Products and Orthogonality

9. Orthogonal complements

slide 28 · Bridge · Companion review, Q9

Problem.

Find a basis for the set of all vectors in $\mathbb{R}^4$ orthogonal to $(1, 1, 1, 1)^\top$ and $(1, 2, 3, 4)^\top$.

Solution.

We want $\ker(\mathbf{A})$ for

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \end{bmatrix}.$$

Free variables x_3, x_4 : $x_1 = x_3 + 2x_4$, $x_2 = -2x_3 - 3x_4$. A basis is

$$\{(1, -2, 1, 0)^\top, (2, -3, 0, 1)^\top\}, \quad \dim = 2.$$

Remarks.

Note. The general fact: the orthogonal complement of $\text{im}(\mathbf{A}^\top)$ is $\ker(\mathbf{A})$. Two independent constraints in $\mathbb{R}^4$ leave a 2-dimensional solution set.

10. Gram–Schmidt by hand

slides 30–31 · Bridge · Companion review, Q11

Problem.

Apply Gram–Schmidt to

$$\mathbf{a} = (1, -1, 0, 0)^\top, \quad \mathbf{b} = (0, 1, -1, 0)^\top, \quad \mathbf{c} = (0, 0, 1, -1)^\top.$$

Then show both $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ and the orthogonalized list are bases for the subspace orthogonal to $\mathbf{d} = (1, 1, 1, 1)^\top$.

Solution.

$$\mathbf{a}' = \mathbf{a} = (1, -1, 0, 0)^\top, \|\mathbf{a}'\|^2 = 2.$$

$$\langle \mathbf{a}', \mathbf{b} \rangle = -1, \text{ so } \mathbf{b}' = \mathbf{b} + \frac{1}{2}\mathbf{a}' = (\frac{1}{2}, \frac{1}{2}, -1, 0)^\top, \|\mathbf{b}'\|^2 = \frac{3}{2}.$$

$$\langle \mathbf{a}', \mathbf{c} \rangle = 0 \text{ and } \langle \mathbf{b}', \mathbf{c} \rangle = -1, \text{ so } \mathbf{c}' = \mathbf{c} + \frac{2}{3}\mathbf{b}' = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1)^\top.$$

Each of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ has entries summing to zero, so each is $\perp \mathbf{d}$; they are independent (staircase pattern), and $\dim \mathbf{d}^\perp = 3$. Gram–Schmidt preserves the span, so the primed list is a basis of the same space.

$$11. \ker(\mathbf{A}^\top \mathbf{A}) = \ker(\mathbf{A})$$

slides 34–35 · Bridge

Problem.

Show that $\ker(\mathbf{A}^\top \mathbf{A}) = \ker(\mathbf{A})$ for any $\mathbf{A} \in \mathbb{R}^{n \times p}$, and deduce that $\text{Rank}(\mathbf{A}^\top \mathbf{A}) = \text{Rank}(\mathbf{A})$.

Then: for which $\mathbf{X}$ is $\mathbf{X}^\top \mathbf{X}$ invertible?

Solution.

$$(\supseteq) \text{ If } \mathbf{A}\mathbf{x} = \mathbf{0} \text{ then } \mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{0}.$$

$$(\subseteq) \text{ If } \mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{0}, \text{ take the inner product with } \mathbf{x}:$$

$$0 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A}\mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \implies \mathbf{A}\mathbf{x} = \mathbf{0}.$$

Both are subspaces of $\mathbb{R}^p$, so rank–nullity gives equal ranks.

Remarks.

Note. Hence $\mathbf{X}^\top \mathbf{X}$ (a $p \times p$ matrix) is invertible **iff $\mathbf{X}$ has full column rank**, i.e. $\ker(\mathbf{X}) = \{\mathbf{0}\}$ — no exact collinearity among the columns. This is exactly the condition the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$ will need later.

Special Matrices and Scalars

12. Two one-line proofs

slide 38 · Bridge · Companion review, Q13 and Q14

Problem.

(a) If $\mathbf{X}$ is any $n \times p$ matrix, show $\mathbf{X}^\top \mathbf{X}$ is symmetric.

(b) For any symmetric non-singular $\mathbf{A}$, show $\mathbf{A}^{-1} = (\mathbf{A}^{-1})^\top$.

Solution.

$$(a) (\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top (\mathbf{X}^\top)^\top = \mathbf{X}^\top \mathbf{X}.$$

$$(b) (\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1} = \mathbf{A}^{-1}, \text{ using symmetry in the last step.}$$

For (b), the identity $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1}$ itself follows from transposing $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$: $(\mathbf{A}^{-1})^\top \mathbf{A}^\top = \mathbf{I}$.

13. The cyclic property

slide 40 · Bridge · Companion review, Q3

Problem.

Prove $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ for all conformable $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{B} \in \mathbb{R}^{m \times n}$.

Solution.

$$\text{tr}(\mathbf{AB}) = \sum_{i=1}^n [\mathbf{AB}]_{ii} = \sum_{i=1}^n \sum_{j=1}^m a_{ij} b_{ji} = \sum_{j=1}^m \sum_{i=1}^n b_{ji} a_{ij} = \sum_{j=1}^m [\mathbf{BA}]_{jj} = \text{tr}(\mathbf{BA}).$$

The whole proof is *swap the order of a finite double sum*. Note $\mathbf{AB}$ is $n \times n$ and $\mathbf{BA}$ is $m \times m$ — different sizes, same trace.

14. Choosing a method

slide 42 · Baseline · Companion review, Q20 and Q21

Problem.

- (a) Compute $\det \mathbf{A}$ for the 4×4 matrix with ones in positions $(1, 4), (2, 1), (3, 2), (4, 3)$ and zeros elsewhere. Which method would you use, and why?
- (b) Using row and column operations, compute

$$\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{vmatrix}, \quad \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}, \quad \begin{vmatrix} 1 & x \\ 1 & y \end{vmatrix}$$

Solution.

- (a) $\det = -1$. Cofactor expansion along row 1 hits a single nonzero entry with sign $(-1)^{1+4}$ and an identity minor. Better still: it is the permutation matrix of the 4-cycle $(1\ 4\ 3\ 2)$, and a k -cycle has sign $(-1)^{k-1}$.
- (b) -12 ; 0 (rows in arithmetic progression, so $R_1 - 2R_2 + R_3 = \mathbf{0}$); $y - x$.

Remarks.

Watch out. Never compute a determinant by brute-force cofactor expansion for $n > 3$: it costs $O(n!)$. Use elimination ($\det = \pm$ product of pivots, $O(n^3)$), or eigenvalues, or structure.

Projections

15. Projecting a vector

slides 48–50 · Bridge · Companion review, Q10

Problem.

Find the orthogonal projection of $(1, 1, 1, 1)^\top$ onto the subspace of $\mathbb{R}^4$ spanned by $\mathbf{v}_1 = (1, 3, 1, 1)^\top$ and $\mathbf{v}_2 = (2, 1, 1, 0)^\top$.

Solution.

Put the spanning vectors in the columns of $\mathbf{A} = [\mathbf{v}_1 \ \mathbf{v}_2]$ and set up the normal equations $\mathbf{A}^\top \mathbf{A} \hat{\mathbf{c}} = \mathbf{A}^\top \mathbf{y}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 12 & 6 \\ 6 & 6 \end{bmatrix}, \quad \mathbf{A}^\top \mathbf{y} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

Solve for the coefficients — $\det(\mathbf{A}^\top \mathbf{A}) = 72 - 36 = 36$:

$$\begin{aligned} \hat{\mathbf{c}} &= (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{y} = \frac{1}{36} \begin{bmatrix} 6 & -6 \\ -6 & 12 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} \\ &= \frac{1}{36} \begin{bmatrix} 36 - 24 \\ -36 + 48 \end{bmatrix} = \frac{1}{36} \begin{bmatrix} 12 \\ 12 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}. \end{aligned}$$

$\hat{\mathbf{c}}$ holds **coefficients**, not the answer. Push them back through $\mathbf{A}$ to land in $\mathbb{R}^4$:

$$\begin{aligned} \mathbf{P}\mathbf{y} &= \mathbf{A}\hat{\mathbf{c}} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix} = \frac{1}{3} \mathbf{v}_1 + \frac{1}{3} \mathbf{v}_2 \\ &= \frac{1}{3} \begin{bmatrix} 1+2 \\ 3+1 \\ 1+1 \\ 1+0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 \\ 4 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4/3 \\ 2/3 \\ 1/3 \end{bmatrix}. \end{aligned}$$

Remarks.

Note. Check by verifying the residual is orthogonal to the plane. With $\mathbf{e} = \mathbf{y} - \mathbf{P}\mathbf{y} = (0, -\frac{1}{3}, \frac{1}{3}, \frac{2}{3})^\top$:

$$\langle \mathbf{e}, \mathbf{v}_1 \rangle = -1 + \frac{1}{3} + \frac{2}{3} = 0, \quad \langle \mathbf{e}, \mathbf{v}_2 \rangle = -\frac{1}{3} + \frac{1}{3} = 0. \checkmark$$

Eigenvalues, Eigenvectors, and Diagonalization

16. Eigen-computations

slides 53–54 · Bridge · Companion review, Q23

Problem.

Find the characteristic polynomials, eigenvalues, and eigenvectors of

$$\begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}, \quad \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}.$$

Solution.

First. $\text{tr} = 1$, $\det = -2$, so $\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1)$. Eigenvalues $2, -1$ with eigenvectors $(5, 2)^\top$ and $(1, 1)^\top$.

Second. $\text{tr} = -3$, $\det = 4$, sum of 2×2 principal minors $= 0$, so

$$p(\lambda) = \lambda^3 + 3\lambda^2 - 4 = (\lambda - 1)(\lambda + 2)^2.$$

$\lambda = 1$: eigenvector $(1, -1, 1)^\top$. $\lambda = -2$: $\mathbf{A} + 2\mathbf{I}$ has rank 1 with null space $\{x + y + z = 0\}$, giving a **two**-dimensional eigenspace, e.g. $(1, -1, 0)^\top, (1, 0, -1)^\top$.

Remarks.

Note. Hold on to that second matrix. Its eigenvalue -2 has algebraic multiplicity 2 *and* geometric multiplicity 2 — which is exactly the condition for diagonalizability, coming up next.

17. Trace, projections, and the spectral theorem

slides 60–63 · Bridge

Problem.

- Prove that $\text{tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i$ for symmetric $\mathbf{A}$.
- What are the possible eigenvalues of a projection matrix? For an orthogonal projector of rank r , what are their multiplicities?
- Find the orthogonal projection matrix onto the one-dimensional subspace of $\mathbb{R}^n$ spanned by $\mathbf{1} = (1, \dots, 1)^\top$.

Solution.

- Write $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ with $\mathbf{U}$ orthogonal, then use the cyclic property of the trace:

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top) = \text{tr}(\mathbf{\Lambda}\mathbf{U}^\top\mathbf{U}) = \text{tr}(\mathbf{\Lambda}) = \sum_{i=1}^n \lambda_i.$$

The only fact used was $\mathbf{U}^\top\mathbf{U} = \mathbf{I}$ — that the two outer factors cancel. The same two lines therefore give $\text{tr}(\mathbf{A}) = \sum_i \lambda_i$ for **any** diagonalizable $\mathbf{A}$, symmetric or not, since there the outer factors are inverses of each other.

- Idempotence pins the eigenvalues down completely. If $\mathbf{P}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then

$$\lambda\mathbf{v} = \mathbf{P}\mathbf{v} = \mathbf{P}^2\mathbf{v} = \lambda^2\mathbf{v} \implies \lambda^2 = \lambda \implies \lambda \in \{0, 1\}.$$

For an orthogonal projector of rank r : $\lambda = 1$ with multiplicity r , with eigenvectors spanning $\text{im}(\mathbf{P})$, and $\lambda = 0$ with multiplicity $n - r$, spanning $\text{ker}(\mathbf{P})$.

- Projecting onto the line spanned by a vector $\mathbf{v}$ gives $\mathbf{P} = \mathbf{v}\mathbf{v}^\top / (\mathbf{v}^\top\mathbf{v})$. Here $\mathbf{v} = \mathbf{1}$ and $\mathbf{1}^\top\mathbf{1} = n$, so

$$\mathbf{P} = \frac{\mathbf{1}\mathbf{1}^\top}{n} = \frac{1}{n}\mathbf{J}, \quad \mathbf{J} = \mathbf{1}\mathbf{1}^\top = \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix}.$$

Check idempotence: $\mathbf{P}^2 = \frac{\mathbf{1}(\mathbf{1}^\top \mathbf{1})\mathbf{1}^\top}{n^2} = \frac{n \mathbf{1}\mathbf{1}^\top}{n^2} = \mathbf{P}$, and $\mathbf{P}$ is symmetric. ✓

Remarks.

Note. Put (a) and (b) together: $\text{tr}(\mathbf{P}) = \sum_i \lambda_i = r = \text{Rank}(\mathbf{P})$. For projections, trace is rank — which is why $\text{tr}(\mathbf{H}) = p$ counts model degrees of freedom.

Note. $\mathbf{P}$ has rank 1, so by (a) and (b) $\text{tr}(\mathbf{P}) = 1$ — and indeed the diagonal holds n copies of $1/n$. Applied to data, $\mathbf{P}\mathbf{y} = \bar{y} \mathbf{1}$: projecting onto $\mathbf{1}$ replaces every entry by the sample mean, and $\mathbf{I} - \mathbf{P}$ centers the data.

Positive Definiteness

18. PSD and the spectral theorem

slide 66 · Bridge · Stage 1, Q8b–c

Problem.

- (a) Prove that $\mathbf{X}^\top \mathbf{X}$ is symmetric positive semi-definite for any real $\mathbf{X}$.
- (b) State the spectral theorem for symmetric $\mathbf{A}$ and use it to characterize when $\mathbf{A}$ is positive semi-definite.

Solution.

- (a) Symmetry: $(\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top \mathbf{X}$. PSD: $\mathbf{z}^\top \mathbf{X}^\top \mathbf{X} \mathbf{z} = \|\mathbf{X} \mathbf{z}\|^2 \geq 0$.
- (b) Write $\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$ with $\mathbf{U}$ orthogonal. Substituting $\mathbf{w} = \mathbf{U}^\top \mathbf{x}$ (a bijection of $\mathbb{R}^n$),

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{w}^\top \mathbf{\Lambda} \mathbf{w} = \sum_{i=1}^n \lambda_i w_i^2.$$

This is ≥ 0 for all $\mathbf{w}$ iff every $\lambda_i \geq 0$ (take $\mathbf{w} = \mathbf{e}_i$ for necessity). So **PSD** $\iff$ **all eigenvalues non-negative**.

19. The covariance matrix is PSD

slides 67–71 · Bridge

Problem.

Let $\mathbf{x} \in \mathbb{R}^p$ be a random vector with $\mathbb{E}[\mathbf{x}] = \mu$.

- (a) Define $\text{Cov}(\mathbf{x})$ two ways: as a single matrix expectation, and entry by entry.
- (b) Prove that $\text{Cov}(\mathbf{x})$ is symmetric and positive semi-definite.

Solution.

- (a) With $\mathbf{z} = \mathbf{x} - \mu$,

$$\Sigma \triangleq \text{Cov}(\mathbf{x}) = \mathbb{E}[\mathbf{z}\mathbf{z}^\top], \quad \Sigma_{ij} = \text{Cov}(x_i, x_j) = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)].$$

The diagonal holds the variances $\text{Var}(x_i)$.

(b) Symmetry is immediate: $\Sigma^\top = \mathbb{E}[(\mathbf{z}\mathbf{z}^\top)^\top] = \mathbb{E}[\mathbf{z}\mathbf{z}^\top] = \Sigma$.

For positive semi-definiteness, fix an **arbitrary** $\mathbf{a} \in \mathbb{R}^p$ and use linearity of expectation to move it inside:

$$\mathbf{a}^\top \Sigma \mathbf{a} = \mathbf{a}^\top \mathbb{E}[\mathbf{z}\mathbf{z}^\top] \mathbf{a} = \mathbb{E}[\mathbf{a}^\top \mathbf{z}\mathbf{z}^\top \mathbf{a}].$$

Now define the **scalar** random variable $Y = \mathbf{a}^\top \mathbf{z}$. Because a scalar equals its own transpose, $\mathbf{a}^\top \mathbf{z} = \mathbf{z}^\top \mathbf{a}$, so the quantity inside the expectation is $Y \cdot Y$:

$$\mathbf{a}^\top \Sigma \mathbf{a} = \mathbb{E}[Y^2].$$

(b, continued) $Y^2 \geq 0$ for every outcome, and the expected value of a non-negative random variable is non-negative. Therefore

$$\mathbf{a}^\top \Sigma \mathbf{a} = \mathbb{E}[Y^2] \geq 0 \quad \text{for every } \mathbf{a} \in \mathbb{R}^p,$$

which is exactly the statement $\Sigma \succeq 0$. ■

Remarks.

Note. Since $\mathbb{E}[Y] = \mathbf{a}^\top \mathbb{E}[\mathbf{z}] = 0$, we also have $\mathbb{E}[Y^2] = \text{Var}(Y) = \text{Var}(\mathbf{a}^\top \mathbf{x})$. So the quadratic form is not merely non-negative — it **is** the variance of the scalar $\mathbf{a}^\top \mathbf{x}$.

Note. Same shape as the $\mathbf{X}^\top \mathbf{X}$ exercise: there $\mathbf{X}^\top \mathbf{X} \succeq 0$ because it is of the form $\mathbf{B}^\top \mathbf{B}$; here $\Sigma \succeq 0$ because it is $\mathbb{E}[\mathbf{z}\mathbf{z}^\top]$ — the population version of the same statement. Given n observations, the sample covariance $\mathbf{S} = \frac{1}{n-1} \mathbf{X}_c^\top \mathbf{X}_c$ is *literally* of the form $\mathbf{B}^\top \mathbf{B}$, so it inherits PSD directly — and like Σ it is $p \times p$.

Block Matrices

20. Block-triangular determinant

slide 80 · Challenge · Companion review, Q22

Problem.

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$ arbitrary, and

$$\mathbf{Y} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{I}_p \end{bmatrix}.$$

Prove $\det \mathbf{Y} = \det \mathbf{A}$.

Solution.

Fastest route: $\mathbf{D} = \mathbf{I}_p$ is invertible and $\mathbf{C} = \mathbf{0}$, so

$$\det \mathbf{Y} = \det(\mathbf{I}_p) \det(\mathbf{A} - \mathbf{B}\mathbf{I}_p^{-1}\mathbf{0}) = \det \mathbf{A}.$$

Without machinery: factor $\mathbf{Y} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_p \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{0} & \mathbf{I}_p \end{bmatrix}$ when $\mathbf{A}$ is invertible; the second factor is unit upper-triangular with determinant 1. Both sides are polynomials in the entries of $\mathbf{A}$, so equality extends to singular $\mathbf{A}$ by continuity.